

PhD Math Camp

Linear Algebra, Regression First

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Outline

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Solving Two Equations

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Matrices

Solving $A\mathbf{x} = \mathbf{b}$ by Elimination

Transpose and Inverse

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The Inverse by Elimination

The Determinant

Determinant, Pivots, and Rank

$A\mathbf{x} = \mathbf{b}$, Answered

Regression in Matrix Form

Fitting a Line

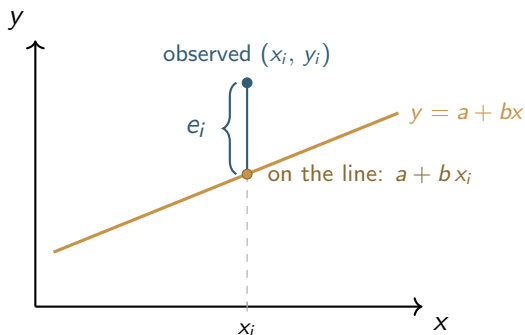
Two Variables, One Line

- We observe n pairs $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.
- We believe y moves with x roughly along a straight line:

$$y \approx a + bx.$$

- From Unit 1.2: b is the *slope*, how much y moves when x moves by one; a is where the line sits when $x = 0$.
- **Which line?** Every pair (a, b) is a different line, and there are infinitely many pairs.

The Residual



Residual

The amount the line misses observation i , measured vertically:

$$e_i = y_i - (a + b x_i).$$

It depends on (a, b) : change the line, and every residual changes.

Why Not Just Add the Misses?

- $\sum_i e_i$ can be zero for a terrible line: misses above the line cancel misses below it.
- $\sum_i |e_i|$ fixes the cancelling, but $|e|$ has no derivative at 0. The tools of Unit 2 stop working.
- $\sum_i e_i^2$: never negative, so nothing cancels. Big misses are punished more than small ones. And it is differentiable.

Square them, then add.

The Sum of Squared Residuals

Sum of squared residuals

$$\text{SSR}(a, b) = \sum_{i=1}^n (y_i - a - b x_i)^2$$

- The data (x_i, y_i) are fixed numbers. The *variables* are a and b .
- So SSR is a function $\mathbb{R}^2 \rightarrow \mathbb{R}$: feed it a line, it returns a score. Lower is better.
- The problem: find the (a, b) that makes SSR smallest.

Careful. Unit 2 minimized functions of *one* variable. This one has two.

Minimizing a Function of Two Variables

- Unit 2, one variable: at a minimum of f , $f'(x^*) = 0$.
- Hold b fixed. Then SSR is a function of a alone, and its derivative is the partial derivative $\partial\text{SSR}/\partial a$.
- Hold a fixed instead: the derivative in b is $\partial\text{SSR}/\partial b$.
- At a minimum, *neither* can be nonzero. If one were, moving a alone (or b alone) would lower SSR further.

First-order condition, two variables

Set every partial derivative to zero, at the same time.

The First-Order Condition

$$\frac{\partial \text{SSR}}{\partial a} = 0, \quad \frac{\partial \text{SSR}}{\partial b} = 0.$$

- Two equations, two unknowns.
- As in Unit 2, the FOC finds *stationary* points. A stationary point could be a minimum, a maximum, or neither.
- In one variable we checked the second derivative. We will not do the two-variable version of that here. Instead: an argument about shape.

Why the FOC Gives a Minimum

- Each term $e_i^2 = (y_i - a - b x_i)^2$ is the *square* of an expression that is linear in (a, b) .
- The square function is convex (Unit 2). Squaring a linear expression keeps it convex.
- A sum of convex functions is convex. So SSR is convex: it is shaped like a bowl.
- A bowl has one bottom. The point where both partials vanish is that bottom, and it is the *global* minimum.

Assume. We take this as given for the rest of the unit.

Proof: Companion Note, §1, Proposition 1.5 and Theorem 1.6.

Taking the Derivatives

Chain rule on each term; the inner derivative in a is -1 , in b it is $-x_i$.

$$\frac{\partial \text{SSR}}{\partial a} = \sum_{i=1}^n 2(y_i - a - b x_i)(-1) = -2 \sum_{i=1}^n e_i$$

$$\frac{\partial \text{SSR}}{\partial b} = \sum_{i=1}^n 2(y_i - a - b x_i)(-x_i) = -2 \sum_{i=1}^n x_i e_i$$

Set both to zero. The -2 cancels:

$$\sum_{i=1}^n e_i = 0, \quad \sum_{i=1}^n x_i e_i = 0.$$

The Normal Equations

Write out e_i and move the unknowns to one side.

$$\sum_i (y_i - a - b x_i) = 0 \implies \sum_i y_i = n a + b \sum_i x_i$$

$$\sum_i x_i (y_i - a - b x_i) = 0 \implies \sum_i x_i y_i = a \sum_i x_i + b \sum_i x_i^2$$

Normal equations

The two equations above. The unknowns are a and b . Everything else is a number computed from the data.

Solving in General

From the first normal equation, dividing by n :

$$a = \bar{y} - b\bar{x}.$$

Substitute into the second and solve for b :

$$b = \frac{\sum_i x_i y_i - n \bar{x} \bar{y}}{\sum_i x_i^2 - n \bar{x}^2} = \frac{S_{xy}}{S_{xx}}, \quad a = \bar{y} - b\bar{x}.$$

- These two lines are all of simple regression.
- We will not use them next. We will solve the two equations directly, to see what kind of problem they are.

Proof: Companion Note, §2, Proposition 2.3.

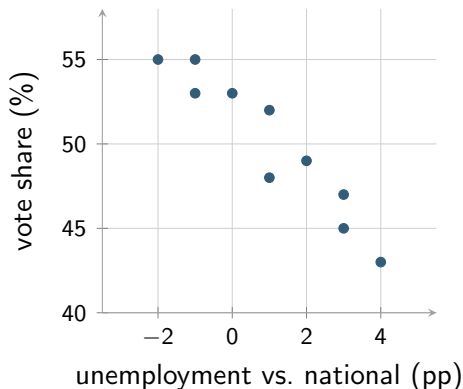
The Data

Does Unemployment Hurt the Incumbent?

- **Retrospective voting.** Voters punish the incumbent party when the economy is bad.
- x_i : unemployment in state i , *relative to the national rate* (percentage points above it, so $x_i < 0$ means the state is doing better than the country).
 y_i : the incumbent party's vote share (%) in state i .
- If the hypothesis is right, the fitted slope b is negative.
- Ten states. The numbers are illustrative, chosen so the arithmetic closes by hand.

Ten States

State	Unempl. x_i	Vote y_i
1	-2	55
2	-1	55
3	-1	53
4	0	53
5	1	52
6	1	48
7	2	49
8	3	45
9	3	47
10	4	43



Which line? We now have a rule: the one that minimizes $\text{SSR}(a, b)$.

The Sums the Normal Equations Need

	1	2	3	4	5	6	7	8	9	10	Σ
x_i	-2	-1	-1	0	1	1	2	3	3	4	10
y_i	55	55	53	53	52	48	49	45	47	43	500
x_i^2	4	1	1	0	1	1	4	9	9	16	46
$x_i y_i$	-110	-55	-53	0	52	48	98	135	141	172	428

$$n = 10, \quad \sum x_i = 10, \quad \sum y_i = 500,$$
$$\sum x_i^2 = 46, \quad \sum x_i y_i = 428.$$

Key Takeaway: Two Equations, Two Unknowns

Plug the sums into the normal equations:

$$500 = 10 a + 10 b$$

$$428 = 10 a + 46 b$$

The calculus is over

What remains is a *system of two linear equations* in two unknowns. Solving the regression means solving this system.

Next: how do we think about solving it, and how many answers can it have?

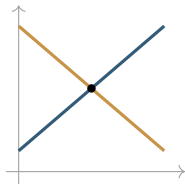
Solving Two Equations

An Equation Is a Line

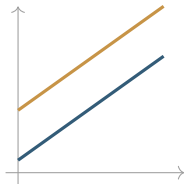
- Take the first equation, $10a + 10b = 500$. Solve for a :
 $a = 50 - b$.
- Every pair (b, a) that satisfies it lies on a line in the (b, a) plane.
- The second, $10a + 46b = 428$, is another line:
 $a = 42.8 - 4.6b$.
- A *solution of the system* is a point that lies on both lines.

So the question “how many solutions?” is the question “how can two lines meet?”

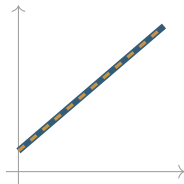
Three Things Two Lines Can Do



They cross.
One solution.



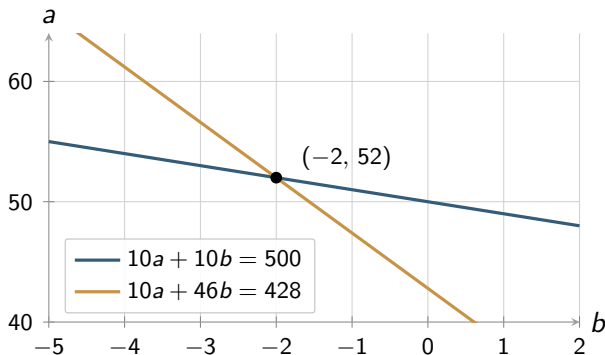
Parallel.
No solution.



The same line.
Infinitely many.

- Parallel: same slope, different intercept. One equation contradicts the other.
- Same line: one equation is a multiple of the other. It adds nothing.

Which Case Are We In?



The slopes differ (-1 and -4.6), so the lines cross once. **Exactly one solution.**

Solving by Elimination

Use the first equation to remove a from the second.

$$10a + 10b = 500 \implies a = 50 - b$$

$$10(50 - b) + 46b = 428$$

$$500 - 10b + 46b = 428$$

$$36b = -72$$

$$b = -2$$

$$a = 50 - (-2) = 52$$

$$\hat{y} = 52 - 2x$$

Check Against the Formulas

$$\bar{x} = 1, \quad \bar{y} = 50,$$

$$S_{xx} = 46 - 10 \cdot 1^2 = 36, \quad S_{xy} = 428 - 10 \cdot 1 \cdot 50 = -72$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{-72}{36} = -2, \quad a = \bar{y} - b\bar{x} = 50 + 2 = 52.$$

- Same answer. The formula and the elimination are the same computation in a different order.
- **Reading.** A state one point worse than the national unemployment rate gives the incumbent two points less of the vote. At the national rate, $a = 52$. The hypothesis survives, in this sample.

Key Takeaway: The Fitted Line

x_i	-2	-1	-1	0	1	1	2	3	3	4
$\hat{y}_i = 52 - 2x_i$	56	54	54	52	50	50	48	46	46	44
$e_i = y_i - \hat{y}_i$	-1	1	-1	1	2	-2	1	-1	1	-1

$$\sum e_i = 0, \quad \sum x_i e_i = 0, \quad \text{SSR} = 16.$$

- The two identities are the FOC, verified on the data.
- Keep $\sum x_i e_i = 0$ in mind. It comes back in Section 8 with a geometric name.

What if there is more than one predictor?

More Than One Predictor

The Linear Model With k Predictors

$$y = a + b_1x_1 + b_2x_2 + \cdots + b_kx_k$$

- Unemployment *and* inflation *and* incumbent tenure, say.
- b_j is the slope in x_j , holding every other predictor fixed (Unit 1.2, and Data Analysis I onward).
- Now $k + 1$ unknowns: $a, b_1, \dots, b_k$.

The FOC, Again

$$\text{SSR}(a, b_1, \dots, b_k) = \sum_{i=1}^n (y_i - a - b_1 x_{i1} - \dots - b_k x_{ik})^2$$

One partial derivative per unknown, each set to zero: $k + 1$ equations. For $k = 2$:

$$\begin{aligned}\sum y_i &= n a + b_1 \sum x_{i1} + b_2 \sum x_{i2} \\ \sum x_{i1} y_i &= a \sum x_{i1} + b_1 \sum x_{i1}^2 + b_2 \sum x_{i1} x_{i2} \\ \sum x_{i2} y_i &= a \sum x_{i2} + b_1 \sum x_{i1} x_{i2} + b_2 \sum x_{i2}^2\end{aligned}$$

Same structure as before: sums from the data times unknowns, one equation per unknown. Still *linear* in the unknowns.

We Are Stuck

- Three equations by substitution is tedious. Eleven is hopeless.
- We saw three possible outcomes for two lines. What are the outcomes for eleven equations, and how would we tell which one we are in?
- We need three things:
 - a *notation* that separates the coefficients from the unknowns,
 - a *procedure* that solves any size of system,
 - a *criterion* for when the solution exists and is unique.

That is what linear algebra is for. Set regression aside for a while.

Matrices

A General Linear System

m equations, n unknowns $x_1, \dots, x_n$:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots \qquad \qquad \qquad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

- a_{ij} : the coefficient in equation i on unknown j . Row first, column second.
- For now the x 's are *unknowns*, not data. The regression unknowns (a, b) will be renamed to fit this later.

Strip Away the Unknowns

The coefficients, the unknowns, and the right-hand sides, each kept together:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Def. (Matrix) A rectangular array of numbers with m rows and n columns; we say A is $m \times n$. Its entries are a_{ij} .

Def. (Vector, for now) A matrix with one column. $\mathbf{x}$ is $n \times 1$, $\mathbf{b}$ is $m \times 1$. Written bold, lowercase.

Notation

$A_{m \times n}$ a matrix with m rows, n columns

a_{ij} the entry of A in row i , column j

$\mathbf{x}, \mathbf{b}, \mathbf{v}$ vectors (one column); bold to tell them from scalars

x_i the i -th entry of $\mathbf{x}$

$\mathbb{R}^n$ the set of all vectors with n real entries

$\mathbf{b}$ versus b bold $\mathbf{b}$ is the right-hand side vector; plain b is a slope

The last line matters for this deck: $y = a + bx$ uses a scalar b , and $A\mathbf{x} = \mathbf{b}$ uses a vector $\mathbf{b}$. They are different objects.

The Dot Product, as a Recipe

Dot product

For two vectors of the same length n ,

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \sum_{i=1}^n u_i v_i.$$

- Multiply matching entries, add up. The result is a *number*.
- Example: $(3, 4) \cdot (2, 1) = 3 \cdot 2 + 4 \cdot 1 = 10$.
- Only defined when the lengths match.

This is the recipe. What the number *means* waits for Section 8.

$A\mathbf{x}$, Row by Row

Matrix times vector

For A $m \times n$ and $\mathbf{x}$ in $\mathbb{R}^n$, $A\mathbf{x}$ is the vector in $\mathbb{R}^m$ whose i -th entry is

$$(A\mathbf{x})_i = (\text{row } i \text{ of } A) \cdot \mathbf{x}.$$

Then $A\mathbf{x} = \mathbf{b}$ says: for every i , row i dotted with $\mathbf{x}$ equals b_i . That is exactly equation i of the system.

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \implies A\mathbf{x} = \begin{pmatrix} 2 \cdot 2 + 1 \cdot 3 \\ 0 \cdot 2 + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$$

$A\mathbf{x}$, Column by Column

The same product, read the other way:

$$A\mathbf{x} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$$

- $A\mathbf{x}$ weights each *column* of A by the matching entry of $\mathbf{x}$, then adds.
- So $A\mathbf{x} = \mathbf{b}$ asks: can $\mathbf{b}$ be built from the columns of A , and with what weights?

Both readings are true at once. Rows for computing; columns for thinking.

A Matrix Is a Function

Fix A . Then $\mathbf{x} \mapsto A\mathbf{x}$ is a function

$$F : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad F(\mathbf{x}) = A\mathbf{x}.$$

Linear transformation

A function F is *linear* if for all vectors $\mathbf{u}, \mathbf{v}$ and scalars c ,

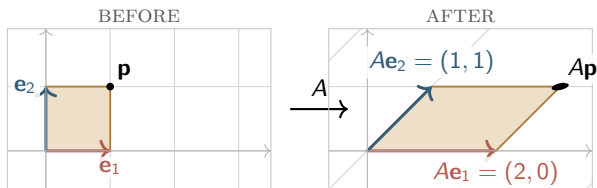
$$F(\mathbf{u} + \mathbf{v}) = F(\mathbf{u}) + F(\mathbf{v}), \quad F(c\mathbf{u}) = cF(\mathbf{u}).$$

- $F(\mathbf{x}) = A\mathbf{x}$ passes both tests, for every A .
- Unit 1.2: $f(x) = bx$ is the one-variable linear function, with one slope. A matrix has one column per input, and each column is that input's slope.

Proof: Companion Note, §4, Proposition 4.1.

What A Does to the Plane

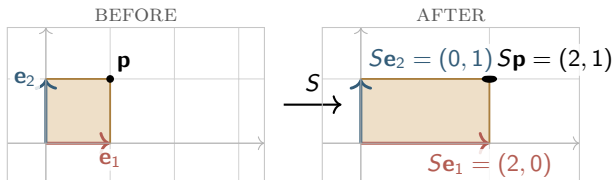
$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ the columns are where $\mathbf{e}_1$ and $\mathbf{e}_2$ land.



- $\mathbf{p} = \mathbf{e}_1 + \mathbf{e}_2$, so by linearity
 $A\mathbf{p} = A\mathbf{e}_1 + A\mathbf{e}_2 = (2, 0) + (1, 1) = (3, 1).$
- Every point follows the same rule. The grid is carried along, and the unit square becomes the parallelogram with the two columns as sides.

Stretching

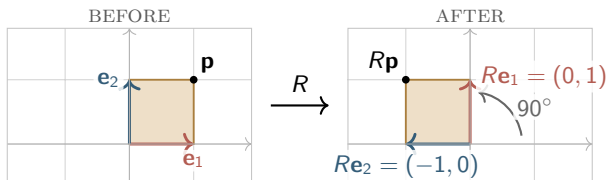
$$S = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \quad S \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ y \end{pmatrix}$$



- Every x -coordinate is doubled; every y -coordinate is left alone. Grid squares become 2×1 rectangles.
- $\mathbf{e}_1$ is stretched to $(2, 0)$; $\mathbf{e}_2$ does not move. Read the columns of S : that is exactly what they say.

Rotating

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad R \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$



- A quarter turn counterclockwise. The square keeps its shape and its size; only its direction changes. Follow the dot on the corner.
- $\mathbf{e}_1$ turns up to $(0, 1)$; $\mathbf{e}_2$ turns left to $(-1, 0)$. Those are the columns of R .

One Transformation After Another

Apply S first, then R : $\mathbf{x} \mapsto R(S\mathbf{x})$.

- The composite is again linear, so it is again a matrix. Call it RS : the matrix of “do S , then do R .”
- Read *right to left*, in the order the transformations act.

Matrix multiplication

For A $m \times n$ and B $n \times p$, the product AB is the $m \times p$ matrix with entries

$$(AB)_{ij} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B).$$

Defined only when the columns of A match the rows of B .

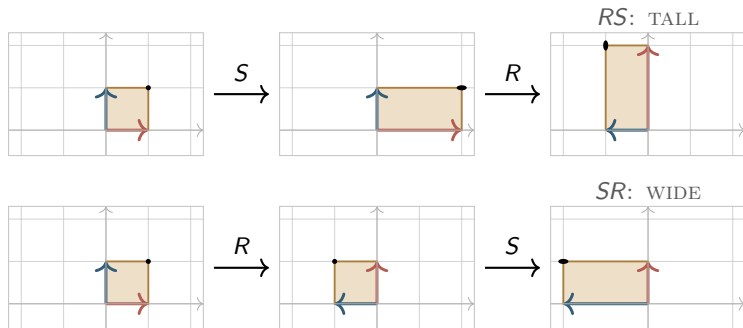
This is the recipe that makes $(AB)\mathbf{x} = A(B\mathbf{x})$ hold.

Proof: Companion Note, §4, Proposition 4.2.

Worked, and a Warning

$$RS = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$$

$$SR = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$$



Same two moves, opposite order, different result. Follow the dot.

Careful. $AB \neq BA$ in general. The order is the order of application, read right to left.

Adding Matrices: Apply Both, Add the Outputs

Entry by entry, and only for matrices of the same shape:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (A - B)_{ij} = a_{ij} - b_{ij}, \quad (cA)_{ij} = c a_{ij}.$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 4 & 6 \end{pmatrix}, \quad 3 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}$$

As transformations:

- $(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}$: send $\mathbf{x}$ through A and through B separately, then add the two outputs.
- $(cA)\mathbf{x} = c(A\mathbf{x})$: scale the output.
- Contrast with AB , which is one transformation *after* the other. Sum: side by side. Product: in sequence.

Rules of Matrix Arithmetic

- (i) $A + B = B + A$, $(A + B) + C = A + (B + C)$
- (ii) $(AB)C = A(BC)$ *(associativity)*
- (iii) $A(B + C) = AB + AC$, $(A + B)C = AC + BC$
(distributivity)
- (iv) $c(AB) = (cA)B = A(cB)$
- (v) $AB \neq BA$ in general *(no commutativity)*
- (vi) $AI = IA = A$, where I is the identity matrix (next slide)

(ii) is the one that is not obvious. It holds because both sides are the matrix of “do C , then B , then A .”

Proof: Companion Note, §4, Corollary 4.3.

Solving $Ax = b$ by Elimination

Back to $A\mathbf{x} = \mathbf{b}$

- We know A and $\mathbf{b}$. We want $\mathbf{x}$.
- In Section 3 we solved a 2×2 system by substitution: use one equation to remove an unknown from the other.
- Turn that into a procedure that works at any size, and that tells us which of the three cases we are in.

The procedure is **Gaussian elimination**. It works on the numbers only, so first we write the system as numbers only.

The Augmented Matrix

$$\begin{array}{rcl} x + y + z & = & 6 \\ x + 2y + 3z & = & 14 \\ 2x + y - z & = & 1 \end{array} \quad \Rightarrow \quad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 1 & -1 & 1 \end{array} \right)$$

Def. (Augmented matrix) $[A \mid \mathbf{b}]$: the coefficient matrix with the right-hand side attached as an extra column. Each row is one equation.

The unknowns are not written. They are the column positions.

Three Row Operations

1. **Swap** two rows.
2. **Scale** a row by a nonzero number.
3. **Add** a multiple of one row to another row.
 - Each one is something legal to do to equations: reorder them, multiply both sides, add one equation to another.
 - So none of them changes the set of solutions.

Row operations preserve the solution set

If $[A' \mid \mathbf{b}']$ is obtained from $[A \mid \mathbf{b}]$ by row operations, then $A'\mathbf{x} = \mathbf{b}'$ and $A\mathbf{x} = \mathbf{b}$ have exactly the same solutions.

Proof: Companion Note, §3, Proposition 3.1.

The Target: Row Echelon Form

$$\left(\begin{array}{ccc|c} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & \blacksquare & * \end{array} \right)$$

- A staircase: each row starts with more zeros than the row above.
- The first nonzero entry in a row is its **pivot**.
- Once there, solve from the bottom up: the last row has one unknown, the one above it two, and so on.

Back-substitution.

The procedure: use row operations to make the entries below each pivot zero, one column at a time, left to right.

Case 1: One Solution

$$\begin{pmatrix} 1 & 1 & 1 & | & 6 \\ 1 & 2 & 3 & | & 14 \\ 2 & 1 & -1 & | & 1 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 2 & | & 8 \\ 0 & -1 & -3 & | & -11 \end{pmatrix}$$
$$\xrightarrow{R_3 + R_2} \begin{pmatrix} 1 & 1 & 1 & | & 6 \\ 0 & 1 & 2 & | & 8 \\ 0 & 0 & -1 & | & -3 \end{pmatrix}$$

Three pivots. Back-substitute:

$$-z = -3 \Rightarrow z = 3, \quad y + 2(3) = 8 \Rightarrow y = 2, \quad x + 2 + 3 = 6 \Rightarrow x = 1.$$

Exactly one solution: $(x, y, z) = (1, 2, 3)$.

Case 2: No Solution

A different coefficient matrix. Note the third row is the sum of the first two on the left, but not on the right.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 3 \\ 2 & 3 & 4 & 6 \end{array} \right) \xrightarrow[\substack{R_2 - R_1 \\ R_3 - 2R_1}]{} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 2 \end{array} \right) \xrightarrow{R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

The last row reads $0x + 0y + 0z = 1$. Nothing satisfies it. **No solution.**
The equations contradict each other.

Case 3: Infinitely Many Solutions

Same coefficients, one right-hand side changed: $6 \rightarrow 5$.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 3 \\ 2 & 3 & 4 & 5 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

The last row is $0 = 0$: true, but it tells us nothing. Two pivots, three unknowns. Set the pivot-less unknown free, $z = t$, and back-substitute:

$$y = 1 - 2t, \quad x = 2 - y - z = 1 + t.$$

$$(x, y, z) = (1 + t, 1 - 2t, t) \quad \text{for any } t \in \mathbb{R}.$$

Infinitely many solutions, one for each t : a line in $\mathbb{R}^3$.

Key Takeaway: What the Last Row Tells You

Echelon form ends with	Meaning	Solutions
a pivot in every column	every unknown is pinned down	exactly one
$(0 \cdots 0 \mid c), c \neq 0$	a contradiction	none
$(0 \cdots 0 \mid 0)$	a free unknown	infinitely many

- Same three cases as two lines in Section 3, now for any size.
- Cases 2 and 3 had the *same* A . The right-hand side decided between none and infinitely many. But A alone decided that “exactly one” was impossible.

Could we have known that from A , before touching $\mathbf{b}$? Sections 11 and 12.

Transpose and Inverse

Transpose and Special Matrices

- **Transpose** A' (also A^T): rows become columns, $(A')_{ij} = a_{ji}$. Turns $m \times n$ into $n \times m$. $(AB)' = B'A'$.
- **Square**: as many rows as columns.
- **Identity** I_n : square, ones on the diagonal, zeros elsewhere. $I\mathbf{x} = \mathbf{x}$ for every $\mathbf{x}$.
- **Diagonal**: square, zero off the diagonal.
- **Symmetric**: $A = A'$. Every matrix of the form $X'X$ is symmetric.
- **Upper triangular**: square, zeros below the diagonal. Row echelon form of a square matrix is upper triangular.
- **Zero matrix** O : all entries zero.

The Inverse, Defined

Inverse of a square matrix

A^{-1} is the $n \times n$ matrix, if one exists, with

$$AA^{-1} = A^{-1}A = I_n.$$

If it exists, A is *invertible* (or *nonsingular*). If not, A is *singular*.

- As a transformation: A^{-1} undoes A . Rotate by 90° , then by -90° : back where you started.
- When it exists, it is unique.
- $(AB)^{-1} = B^{-1}A^{-1}$: undo the last thing first.

Proof: Companion Note, §6, Propositions 6.1 and 6.2.

Two Questions the Inverse Raises

If A^{-1} exists, then $A\mathbf{x} = \mathbf{b}$ is solved in one line:

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

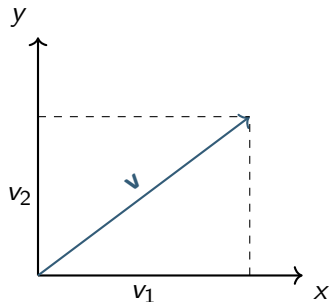
(Section 13 does this carefully.) That leaves two questions.

1. **How do we compute A^{-1} ?** Section 10
2. **When does A^{-1} exist at all?** Sections 11, 12

Both answers use the idea of vectors pointing in “genuinely different” directions. So first, vectors, properly.

Vectors

Vectors as Arrows



- A vector $\mathbf{v} = (v_1, v_2)$ is an arrow from the origin to the point (v_1, v_2) . It has a length and a direction.
- The entries are its horizontal and vertical components. The same column of numbers from Section 5, now drawn.

Vectors as Data: Two Readings

Back to the ten states.

- **A row is a vector.** State 5 is the point $(1, 52)$ in $\mathbb{R}^2$: its relative unemployment and its vote share. The scatter in Section 2 was ten such vectors.
- **A column is a vector.** Unemployment across all ten states is

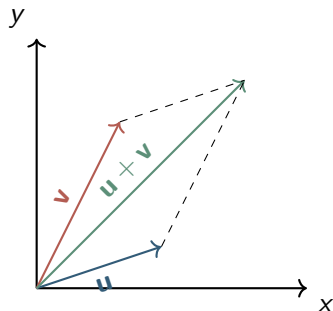
$$\mathbf{x} = (-2, -1, -1, 0, 1, 1, 2, 3, 3, 4) \in \mathbb{R}^{10},$$

and vote share is $\mathbf{y} \in \mathbb{R}^{10}$.

A variable is a vector

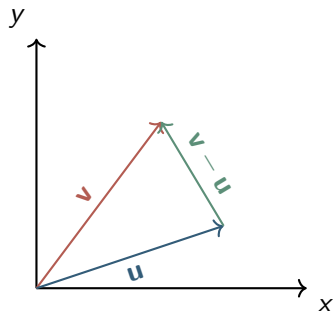
We cannot draw $\mathbb{R}^{10}$, but every operation below works there exactly as in the plane. Correlation and regression live in this reading.

Addition and Subtraction, Drawn



Add: place $\mathbf{v}$ at the tip of $\mathbf{u}$.

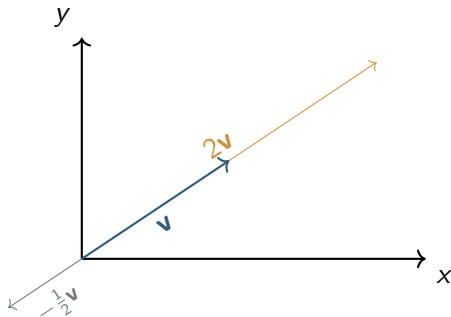
$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix},$$



Subtract: the arrow from the tip of $\mathbf{u}$ to the tip of $\mathbf{v}$.

$$\mathbf{v} - \mathbf{u} = \begin{pmatrix} v_1 - u_1 \\ v_2 - u_2 \end{pmatrix}.$$

Scalar Multiplication

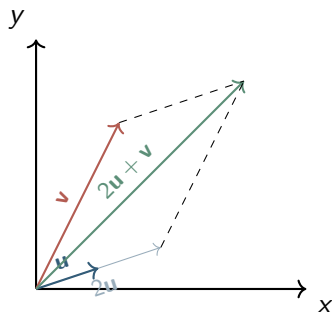


- $c\mathbf{v} = (c v_1, c v_2)$: every entry times c .
- $c > 0$ stretches or shrinks along the same direction. $c < 0$ reverses it.
- All multiples of $\mathbf{v}$ together form a *line* through the origin.

Linear Combinations

Linear combination

$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$, for scalars $c_1, \dots, c_k$.



Scale each vector, then add. $A\mathbf{x}$ read column by column (Section 5) was exactly this: a linear combination of the columns of A with weights $\mathbf{x}$.

Length and Distance

Norm

$\|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}}$, the length of the arrow.

- In the plane this is Pythagoras: $\|(3, 4)\| = \sqrt{9 + 16} = 5$.
- **Distance** between two points is the length of their difference:
 $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.
- The dot product of a vector with itself is its squared length.
The recipe from Section 5 has started to mean something.

$\text{SSR} = \sum e_i^2 = \mathbf{e} \cdot \mathbf{e} = \|\mathbf{e}\|^2$. Least squares makes the residual vector *short*.

The Dot Product, Geometrically

Dot product and angle

For nonzero $\mathbf{u}, \mathbf{v}$ with angle θ between them,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta, \quad \text{so} \quad \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

- Same direction: $\cos \theta = 1$. Opposite: -1 . Perpendicular: 0 .
- In $\mathbb{R}^{10}$ there is no protractor. The formula *is* the definition of the angle there, and it always gives a number in $[-1, 1]$.

Proof: Companion Note, §5, Proposition 5.1 and Theorem 5.2.

Orthogonality and Projection

Orthogonal

$\mathbf{u} \perp \mathbf{v}$ if $\mathbf{u} \cdot \mathbf{v} = 0$. In the plane: a right angle, since $\cos 90^\circ = 0$.

- **Projection** of $\mathbf{u}$ onto $\mathbf{v}$: the piece of $\mathbf{u}$ that points along $\mathbf{v}$,

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

- What is left over, $\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$, is orthogonal to $\mathbf{v}$.
- Example: projecting $(3, 4)$ onto $(2, 0)$ gives $\frac{6}{4}(2, 0) = (3, 0)$; the leftover $(0, 4)$ is perpendicular.

Correlation Is a Cosine

Sample correlation, as it is usually written:

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

Center each variable: $\mathbf{x}_c = \mathbf{x} - \bar{x}\mathbf{1}$, $\mathbf{y}_c = \mathbf{y} - \bar{y}\mathbf{1}$. Then

$$r_{xy} = \frac{\mathbf{x}_c \cdot \mathbf{y}_c}{\|\mathbf{x}_c\| \|\mathbf{y}_c\|} = \cos \theta.$$

What correlation measures

The cosine of the angle between the two centered data vectors in $\mathbb{R}^n$. That is why it lives in $[-1, 1]$.

The Ten States, Again

$$\bar{x} = 1, \bar{y} = 50.$$

$x_i - \bar{x}$	-3	-2	-2	-1	0	0	1	2	2	3
$y_i - \bar{y}$	5	5	3	3	2	-2	-1	-5	-3	-7

$$\mathbf{x}_c \cdot \mathbf{y}_c = -72, \quad \|\mathbf{x}_c\| = \sqrt{36} = 6, \quad \|\mathbf{y}_c\| = \sqrt{160} = 4\sqrt{10}$$

$$r_{xy} = \frac{-72}{6 \cdot 4\sqrt{10}} = -\frac{3}{\sqrt{10}} \approx -0.949$$

- The two centered vectors point almost exactly opposite ways.
- Notice $\mathbf{x}_c \cdot \mathbf{y}_c = S_{xy} = -72$ and $\|\mathbf{x}_c\|^2 = S_{xx} = 36$. The slope was $b = S_{xy}/S_{xx}$: a dot product over a squared length.

Key Takeaway: The FOC Was Orthogonality

Section 1 ended with $\sum_i x_i e_i = 0$. In vector language:

$$\mathbf{x} \cdot \mathbf{e} = 0, \quad \text{that is,} \quad \mathbf{x} \perp \mathbf{e}.$$

- Least squares chooses the line whose residual vector is *orthogonal to the predictor*. (And $\sum e_i = 0$ says $\mathbf{e} \perp \mathbf{1}$.)
- The fitted values $\hat{\mathbf{y}} = a\mathbf{1} + b\mathbf{x}$ are a linear combination of $\mathbf{1}$ and $\mathbf{x}$. So $\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the plane those two vectors span, and $\mathbf{e}$ is the leftover.

That plane is spanned by two vectors. What if one were a multiple of the other? Then they would span only a line, and the projection would not be determined. Next: when do vectors point in genuinely different directions?

Linear Independence

Span

Span

$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set of all linear combinations $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k$: everything you can reach from those vectors by scaling and adding.

- One nonzero vector spans a line.
- Two vectors span a plane, *unless* one is a multiple of the other; then they still only span a line.
- Three vectors in $\mathbb{R}^3$ span all of space, unless they all lie in one plane.

The “unless” is the whole subject of this section.

Intuition Behind Linear Independence

- Imagine vectors as arrows in space; some arrows can be expressed as a combination of others.
- If a vector can be written as a combination of the others, it doesn't add new direction to the space.
- Vectors that each provide a new direction are **linearly independent**. Each one enlarges the span.
- A set with a redundant vector is **linearly dependent**. Removing that vector leaves the span unchanged.

Formal Definition

Linear independence

$\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is *linearly independent* if the only solution to

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0}$$

is $c_1 = c_2 = \dots = c_k = 0$. Otherwise it is *linearly dependent*.

- **Why zero?** If some $c_j \neq 0$ works, divide by it:
 $\mathbf{v}_j = -\frac{1}{c_j} \sum_{i \neq j} c_i \mathbf{v}_i$. So $\mathbf{v}_j$ is built from the others, and is redundant.
- Conversely, if some $\mathbf{v}_j$ is a combination of the others, move it across: a nonzero combination equals $\mathbf{0}$.

Checking It Is Solving a System

Are these independent?

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}$$

$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$ is the system $A\mathbf{c} = \mathbf{0}$ with the $\mathbf{v}$'s as the columns of A . Row-reduce (Section 6, Case 3, same matrix):

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 4 & 0 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

A zero row: a free unknown, so nonzero solutions exist.

Dependent. Indeed $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$; the three vectors span only a plane.

Where You Will Meet This

- **Multicollinearity.** Two predictors that are (nearly) linear combinations of each other. The columns of the data matrix are dependent, and the regression cannot separate their effects.
- Dummy variables for every category *plus* a constant: the dummies add up to the constant column. Dependent by construction. That is why one category is dropped.
- The rank condition on $X'X$ in Section 14 is exactly “the columns of X are independent.”

Now we can answer the inverse's two questions. First: how to compute it.

The Inverse by Elimination

Solve n Systems at Once

$AA^{-1} = I$ says: the j -th column of A^{-1} , call it $\mathbf{x}_j$, solves $A\mathbf{x}_j = \mathbf{e}_j$, where $\mathbf{e}_j$ is the j -th column of I .

- That is n systems, all with the same A .
- Elimination only touches the left side, so do them together: augment A with all of I at once.

The procedure

$$[A \mid I] \xrightarrow{\text{row operations}} [I \mid A^{-1}]$$

Reduce past echelon form until the left block is I . The right block is then A^{-1} .

Worked: $A \ 2 \times 2$ Inverse

$$\begin{aligned} \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) &\xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \right) \xrightarrow{R_2 - 2R_1} \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array} \right) \\ &\xrightarrow{-R_2} \left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array} \right) \xrightarrow{R_1 - R_2} \left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array} \right) \\ A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \quad AA^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \checkmark \end{aligned}$$

When It Fails, and an Exercise

- If elimination produces a **zero row** on the left, the left block can never become I . Then A^{-1} does not exist.
- Section 6's singular matrix is such a case: one row of the left block was a combination of the others, and elimination exposed it.

Ex. (Exercise) Find the inverse of $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & -1 \end{pmatrix}$ (Section 6, Case 1), or show that it has none.

So a zero row is the sign of failure. Is there a single *number* that predicts it?

The Determinant

Where $ad - bc$ Comes From

Run the inverse procedure on a general 2×2 matrix, assuming $a \neq 0$:

$$\left(\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right) \xrightarrow{R_2 - \frac{c}{a}R_1} \left(\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & d - \frac{bc}{a} & -\frac{c}{a} & 1 \end{array} \right)$$

The second pivot is $d - \frac{bc}{a} = \frac{ad-bc}{a}$. To continue we must divide by it.
That is possible only if $ad - bc \neq 0$.

Determinant, 2×2

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

and, when $\det A \neq 0$,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Proof: Companion Note, §6, Proposition 6.4.

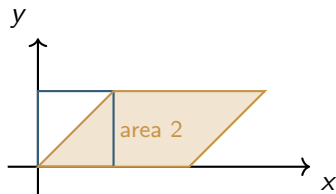
The Determinant as Area

- $|\det A|$ is the *area* of the parallelogram that A makes from the unit square. The sign records whether orientation flipped.

- $A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$: $\det A = 2$. The square of area 1 became a parallelogram of area 2.

- $\det A = 0$: the columns are collinear, the parallelogram is flat, the plane has been squashed onto a line. Nothing can undo that.

$\det A = 0$ and “ A is singular” are the same statement, seen algebraically and geometrically.



Larger Matrices: Cofactor Expansion

Expand along the first row: each entry times the determinant of what is left when its row and column are deleted, with alternating signs.

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Section 6's singular matrix:

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{pmatrix} = 1(8 - 9) - 1(4 - 6) + 1(3 - 4) = -1 + 2 - 1 = 0.$$

Zero, as the zero row predicted. $|\det|$ is now a *volume*: the three columns span a flat box.

Rules for Determinants

For square A, B of the same size:

- (i) Swapping two rows multiplies $\det$ by -1 .
- (ii) Scaling a row by c multiplies $\det$ by c .
- (iii) Adding a multiple of one row to another leaves $\det$ unchanged.
- (iv) A row of zeros, or two equal rows, gives $\det = 0$.
- (v) $\det(AB) = \det A \cdot \det B$.
- (vi) $\det(A') = \det A$; so every rule above also holds for columns.
- (vii) $\det I = 1$, and $\det(A^{-1}) = 1/\det A$ when A is invertible.
- (viii) Triangular A : $\det A$ is the product of the diagonal entries.

(i)–(iii) are the three row operations. That is the link to elimination.

Proof: Companion Note, §7.

Determinant, Pivots, and Rank

Row Operations Track the Determinant

Reduce A to echelon form U . For a square matrix, U is upper triangular, so $\det U$ is the product of its pivots.

- Every row operation changed $\det$ by a known nonzero factor (-1 , or $c \neq 0$, or 1).
- So $\det A = (\text{nonzero factor}) \times (\text{product of pivots})$.
- $\det A \neq 0$ if and only if every pivot is nonzero, if and only if there is no zero row: n pivots for n columns.

Determinant and pivots

For A $n \times n$: $\det A \neq 0 \iff$ elimination produces n pivots
 $\iff A^{-1}$ exists.

Proof: Companion Note, §8, Proposition 8.1 and Theorem 8.2.

Rank

Rank

$\text{rank}(A)$ is the number of pivots in the echelon form of A . Equivalently, the number of linearly independent columns; equivalently, the dimension of the span of the columns.

Rank criterion for $Ax = b$

Let A be $m \times n$.

- $\text{rank}(A) \neq \text{rank}[A \mid \mathbf{b}]$: no solution.
- $\text{rank}(A) = \text{rank}[A \mid \mathbf{b}] = n$: exactly one.
- $\text{rank}(A) = \text{rank}[A \mid \mathbf{b}] < n$: infinitely many.

Section 6 again: Case 1 had $\text{rank } 3 = n$; Cases 2 and 3 had $\text{rank } 2$, and the augmented rank was 3 (contradiction) or 2 (free variable).

Proof: Companion Note, §8, Proposition 8.3.

One Theorem, Six Faces

Invertibility

For a square $n \times n$ matrix A , the following are equivalent:

1. A^{-1} exists (A is nonsingular).
2. $\det A \neq 0$.
3. $\text{rank}(A) = n$ (full rank).
4. The columns of A are linearly independent.
5. Elimination gives n pivots; no zero row.
6. $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b}$.

Every section since 6 has been building one of these faces. They are the same fact.

Proof: Companion Note, §8, Theorem 8.2.

$Ax = b$, Answered

If $\det A \neq 0$, Multiply by the Inverse

$$A\mathbf{x} = \mathbf{b}$$

$$A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{b}$$

$$(A^{-1}A)\mathbf{x} = A^{-1}\mathbf{b}$$

$$I\mathbf{x} = A^{-1}\mathbf{b}$$

$$\mathbf{x} = A^{-1}\mathbf{b}$$

- Each line uses one rule: associativity, then $A^{-1}A = I$, then $I\mathbf{x} = \mathbf{x}$.
- Multiplied on the *left* on both sides. Order matters.
- If $\det A = 0$: no inverse, and no unique solution. Whether there are none or infinitely many depends on $\mathbf{b}$ (Section 6).

The Normal Equations, a Third Time

Section 2's system, as a matrix equation. The unknown vector is (a, b) :

$$\begin{pmatrix} 10 & 10 \\ 10 & 46 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 500 \\ 428 \end{pmatrix}$$

$$\det = 10 \cdot 46 - 10 \cdot 10 = 460 - 100 = 360 \neq 0.$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{360} \begin{pmatrix} 46 & -10 \\ -10 & 10 \end{pmatrix} \begin{pmatrix} 500 \\ 428 \end{pmatrix} = \frac{1}{360} \begin{pmatrix} 18720 \\ -720 \end{pmatrix} = \begin{pmatrix} 52 \\ -2 \end{pmatrix}$$

Same line, $\hat{y} = 52 - 2x$, by a third route.

Key Takeaway: Three Routes, One Answer

Route	Where
Closed-form $b = S_{xy}/S_{xx}$, $a = \bar{y} - b\bar{x}$	Section 3
Elimination on the two equations	Section 3
$\mathbf{x} = A^{-1}\mathbf{b}$	here

- The third route is the only one that says, before computing, whether an answer exists: $\det \neq 0$.
- And it is the only one that scales. Eleven unknowns is the same line, $\mathbf{x} = A^{-1}\mathbf{b}$, with a bigger A .

Back to Section 4's problem, with the notation to write it down.

Regression in Matrix Form

The Data Matrix

n observations, k predictors. One row per observation, one column per predictor, plus a column of ones for the intercept:

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}_{n \times (k+1)}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} a \\ b_1 \\ \vdots \\ b_k \end{pmatrix}$$

- Row i of X dotted with $\boldsymbol{\beta}$ is $a + b_1 x_{i1} + \cdots + b_k x_{ik}$: the prediction for observation i .
- So $X\boldsymbol{\beta}$ is the vector of all n predictions at once.

Residuals as a Vector

$$\mathbf{e} = \mathbf{y} - X\boldsymbol{\beta}, \quad \text{SSR}(\boldsymbol{\beta}) = \sum_{i=1}^n e_i^2 = \mathbf{e} \cdot \mathbf{e} = \mathbf{e}'\mathbf{e}$$

$$\text{SSR}(\boldsymbol{\beta}) = (\mathbf{y} - X\boldsymbol{\beta})'(\mathbf{y} - X\boldsymbol{\beta}).$$

Multiply out, using $(X\boldsymbol{\beta})' = \boldsymbol{\beta}'X'$:

$$\text{SSR}(\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'X'\mathbf{y} + \boldsymbol{\beta}'(X'X)\boldsymbol{\beta}.$$

- A constant, a term linear in $\boldsymbol{\beta}$, and a term quadratic in $\boldsymbol{\beta}$. Same shape as $\text{SSR}(a, b)$ in Section 1, written for $k + 1$ unknowns.
- $X'X$ is $(k + 1) \times (k + 1)$ and symmetric. It will be the coefficient matrix of the normal equations.

The Gradient

Gradient

For $f : \mathbb{R}^{k+1} \rightarrow \mathbb{R}$, the gradient is the vector of all partial derivatives:

$$\nabla f(\boldsymbol{\beta}) = \left(\frac{\partial f}{\partial a}, \frac{\partial f}{\partial b_1}, \dots, \frac{\partial f}{\partial b_k} \right)'.$$

The FOC of Section 1, “set every partial to zero,” is $\nabla SSR = \mathbf{0}$.
Two facts do the differentiation:

$$\nabla_{\boldsymbol{\beta}} (\mathbf{c}'\boldsymbol{\beta}) = \mathbf{c}, \quad \nabla_{\boldsymbol{\beta}} (\boldsymbol{\beta}'M\boldsymbol{\beta}) = 2M\boldsymbol{\beta} \quad (M \text{ symmetric}).$$

They are the vector versions of $\frac{d}{dx}(cx) = c$ and $\frac{d}{dx}(mx^2) = 2mx$.

Proof: Companion Note, §9, Lemmas 9.1 and 9.2.

The Normal Equations in Matrix Form

Apply the two facts to $\text{SSR}(\beta) = \mathbf{y}'\mathbf{y} - 2\beta'X'\mathbf{y} + \beta'(X'X)\beta$:

$$\nabla \text{SSR}(\beta) = -2X'\mathbf{y} + 2X'X\beta.$$

Set to $\mathbf{0}$, cancel the 2:

Normal equations

$$X'X\beta = X'\mathbf{y}$$

- $k + 1$ linear equations in $k + 1$ unknowns. Section 4's pickle, now in one line.
- Equivalently $X'(\mathbf{y} - X\beta) = X'\mathbf{e} = \mathbf{0}$: the residual is orthogonal to every column of X . Section 8, generalized.

Proof: Companion Note, §9, Proposition 9.3.

The Solution

Section 13 says exactly what to do with $A\mathbf{x} = \mathbf{b}$ when A is invertible. Here $A = X'X$, $\mathbf{x} = \boldsymbol{\beta}$, $\mathbf{b} = X'\mathbf{y}$:

Ordinary least squares

$$\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y}, \quad \text{provided } (X'X)^{-1} \text{ exists.}$$

- It is a minimum, not just a stationary point, for the same reason as in Section 1: SSR is a sum of squares, hence convex.
- Every symbol in this line has now been defined in this unit.

Proof: Companion Note, §9, Proposition 9.5.

When Does $(X'X)^{-1}$ Exist?

The rank condition

$X'X$ is invertible if and only if the columns of X are linearly independent, that is, $\text{rank}(X) = k + 1$.

What breaks it:

- Fewer observations than unknowns: $n < k + 1$. At most n independent columns are possible.
- A predictor that is a linear combination of others: perfect multicollinearity. Income in dollars and income in thousands of dollars.
- A full set of category dummies plus the constant column.

When it fails, the normal equations have infinitely many solutions (Section 6, Case 3): the data cannot tell the coefficients apart.

Proof: Companion Note, §9, Theorem 9.4.

The Ten States in Matrix Form

$k = 1$. X is 10×2 : a column of ones and the unemployment column.

$$X'X = \begin{pmatrix} \sum 1 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = \begin{pmatrix} 10 & 10 \\ 10 & 46 \end{pmatrix}, \quad X'y = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \begin{pmatrix} 500 \\ 428 \end{pmatrix}$$

- $X'X \beta = X'y$ is *exactly* the two normal equations of Section 2. The matrix form did not change the problem; it named it.
- $\det(X'X) = 360 \neq 0$: the two columns of X are independent, because relative unemployment is not constant across states.
- $\hat{\beta} = (X'X)^{-1}X'y = (52, -2)$, as in Section 13.

What This Unit Bought You

Idea here	Where it is spent
FOC in several variables	every estimator you will meet
Convexity of a sum of squares	why OLS has one answer
Matrix as coefficients of a system	the normal equations at any k
$A\mathbf{x}$ as a transformation	$X\beta$: predictions from coefficients
Elimination, three cases	what software does to “solve”
Dot product, norm, angle	correlation; residual $\perp$ predictors
Linear independence	multicollinearity, the dummy-variable trap
Determinant, rank, inverse	when $(X'X)^{-1}$ exists at all

The sentence this unit was aimed at

$$\hat{\beta} = (X'X)^{-1}X'y$$

A convex minimization, whose FOC is a linear system, solved by an inverse that exists when the columns of X are independent.